

EXPERIMENT: FARM ANIMALS USING CLASSES AND OBJECTS

Experiment No.	04
Date	20/09/2026
Title	Farm Animals Java Application using Classes and Objects

Objective / Aim

To design and implement a Java application for farm animals such as cow, pig, and horse using classes and objects, and display details such as where each animal stays, what it eats, and the sound it makes.

System Requirements

Hardware

- Standard PC/Laptop
- Minimum 4 GB RAM

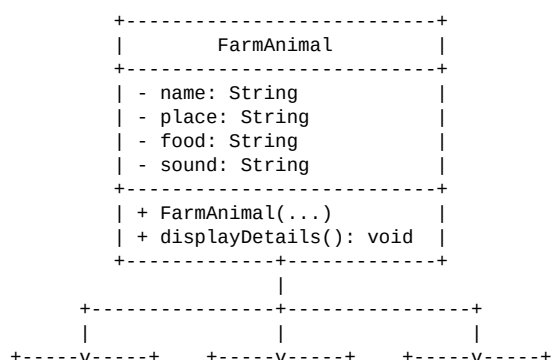
Software

- Windows / Linux / macOS
- Java Development Kit (JDK 17 or later)
- Text Editor or Java IDE
- Command Prompt / Terminal

Algorithm / Logic Description

1. Start the program.
2. Create a FarmAnimal class with attributes for animal name, place, food, and sound.
3. Create a constructor to initialize the animal details.
4. Create a displayDetails() method.
5. Create objects for Cow, Pig, and Horse.
6. Call displayDetails() for each object.
7. Display the animal name, place, food, and sound.
8. Stop the program.

UML Class Diagram



	Cow			Pig			Horse	
+-----+		+-----+		+-----+				

Source Code

```
class FarmAnimal {
    String name;
    String place;
    String food;
    String sound;

    FarmAnimal(String name, String place,
                String food, String sound) {
        this.name = name;
        this.place = place;
        this.food = food;
        this.sound = sound;
    }

    void displayDetails() {
        System.out.println("Animal Name : " + name);
        System.out.println("Stays At      : " + place);
        System.out.println("Eats        : " + food);
        System.out.println("Sound       : " + sound);
        System.out.println();
    }
}

public class FarmAnimals {
    public static void main(String[] args) {

        FarmAnimal cow = new FarmAnimal(
            "Cow", "Cowshed", "Grass and Hay", "Moo");

        FarmAnimal pig = new FarmAnimal(
            "Pig", "Pigsty", "Grains and Vegetables", "Oink");

        FarmAnimal horse = new FarmAnimal(
            "Horse", "Stable", "Grass and Hay", "Neigh");

        System.out.println("==== FARM ANIMALS =====");

        System.out.println("\n--- COW ---");
        cow.displayDetails();

        System.out.println("\n--- PIG ---");
        pig.displayDetails();

        System.out.println("\n--- HORSE ---");
        horse.displayDetails();
    }
}
```

Compilation & Execution Commands

```
javac FarmAnimals.java
java FarmAnimals
```

Sample Output

```
==== FARM ANIMALS =====

--- COW ---
Animal Name : Cow
Stays At    : Cowshed
Eats       : Grass and Hay
Sound      : Moo

--- PIG ---
Animal Name : Pig
Stays At    : Pigsty
Eats       : Grains and Vegetables
Sound      : Oink

--- HORSE ---
Animal Name : Horse
Stays At    : Stable
Eats       : Grass and Hay
Sound      : Neigh
```

Result

The Farm Animals Java application was successfully implemented using classes and objects. The program creates objects for cow, pig, and horse and displays their place of stay, food, and sound.